



Essay / Assignment Title: Choosing machine learning and optimization techniques for real-world business/social problems

Programme title: MSc Data Analytics

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Year: 2026

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CHAPTER ONE: INTRODUCTION

Predictive analytics and machine learning have become essential tools for data-driven decision making. Organizations collect large amounts of data about customers, products, services, transactions, and outcomes, but raw data alone has limited value. Business value is created when data is cleaned, processed, analyzed, and transformed into insights that support better decisions. Predictive analytics helps estimate future outcomes from historical patterns, while machine learning automatically learns relationships between input features and target outcomes.

In business settings, predictive analytics is widely used for customer segmentation, risk prediction, fraud detection, demand forecasting, recommendation systems, and marketing campaign optimization. For example, a bank can use historical campaign data to identify customers who are more likely to accept a marketing offer, instead of contacting customers randomly. This can reduce marketing costs, improve customer targeting, and increase campaign success.

This report focuses on a real-world business classification problem using the UCI Bank Marketing dataset ([Moro, 2014](#)). The dataset is based on direct marketing campaigns of a Portuguese banking institution, where customers were contacted mainly by phone. The goal is to predict whether a customer will subscribe to a bank term deposit. The target variable is y , which indicates whether the customer subscribed or not.

The project applies two machine learning algorithms: Support Vector Machine (SVM) and XGBoost. SVM is a supervised learning method used for classification and is effective in high-dimensional spaces. XGBoost is a scalable tree boosting algorithm widely used for structured/tabular data because of its efficiency and strong predictive performance.

The report follows a complete analytical project lifecycle. It includes dataset description, data type identification, data cleaning, preprocessing, encoding, exploratory data analysis, SVM model training and evaluation, 3D visualization of class separation, XGBoost implementation, hyperparameter tuning, and model comparison using accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrices.

CHAPTER TWO: Dataset Description and Business Problem

The selected dataset is the Bank Marketing dataset from the UCI Machine Learning Repository (Moro, 2014). The dataset contains information about customers, their demographic and financial characteristics, previous marketing contact history, and campaign-related variables. The business problem is to predict whether a customer will subscribe to a term deposit after being contacted by the bank.

This is a binary classification problem because the target variable has two possible outcomes:

Table 1. Target variable data description

Target value	Meaning
no	Customer did not subscribe to a term deposit
yes	Customer subscribed to a term deposit

The machine learning objective is to learn patterns from customer and campaign data and predict the probability of subscription. In a real banking environment, such a model could support marketing managers by helping them identify customers who are more likely to respond positively to a campaign. This would allow the bank to prioritize high-potential customers, reduce unnecessary calls, improve campaign efficiency, and use resources more effectively.

The dataset is multi-dimensional because it contains more than three input attributes. Important features include:

Table 2. Feature description of the bank marketing dataset

Feature	Description	Data type
age	Age of the customer	Numerical
job	Job type of the customer	Categorical

marital	Marital status	Categorical
education	Education level	Ordinal categorical
default	Whether customer has credit default	Binary categorical
balance	Average yearly balance	Numerical
housing	Whether customer has housing loan	Binary categorical
loan	Whether customer has personal loan	Binary categorical
contact	Contact communication type	Categorical
day	Last contact day of the month	Numerical
month	Last contact month	Categorical
duration	Last contact duration	Numerical
campaign	Number of contacts during campaign	Numerical
pdays	Days since previous campaign contact	Numerical
previous	Number of previous contacts	Numerical
poutcome	Previous campaign outcome	Categorical

This dataset is suitable for the assignment because it contains numerical, categorical, binary categorical, and ordinal variables. Therefore, it allows the use of data cleaning, encoding, scaling, exploratory data analysis, classification modelling, model evaluation, and optimization.

CHAPTER THREE: Data Types, Cleaning, and Preprocessing

The dataset contains different types of variables. The numerical columns include variables such as age, balance, day, duration, campaign, pdays, and previous. These variables contain quantitative values and can be used for statistical analysis, correlation analysis, scaling, and model training.

The categorical variables include job, marital, contact, month, and poutcome. These variables do not have a natural numerical order, so one-hot encoding was used for them. One-hot encoding converts each category into a separate binary column. For example, the job variable is transformed into separate columns such as job_management, job_blue-collar, job_student, and others.

The variable education was treated as an ordinal categorical variable because its categories have a natural order: primary < secondary < tertiary. Therefore, ordinal encoding was applied to the education feature. This preserves the ranking structure of education levels. The target variable y was label encoded: no = 0, and yes = 1.

The dataset also contained unknown values in some categorical columns. These values were treated as missing or uncertain information. In preprocessing, unknown values were replaced with missing values, and categorical missing values were imputed using the most frequent category. Numerical missing values were imputed using the median.

Feature scaling was applied to numerical variables using standardization. This step was especially important for SVM because SVM ([Cortes, 1995](#)) is sensitive to the scale of features. Features with larger numerical ranges can dominate the distance calculations if scaling is not applied. Standardization transforms numerical variables so that they have approximately zero mean and unit variance.

The preprocessing pipeline included:

1. Separating features and target variable
2. Replacing unknown values with missing values

3. Encoding the target variable
4. Identifying numerical, nominal categorical, and ordinal categorical columns
5. Applying median imputation and standard scaling to numerical variables
6. Applying ordinal encoding to education
7. Applying one-hot encoding to nominal categorical variables
8. Splitting the data into training and testing sets using stratified sampling

Stratified sampling was used because the target variable was imbalanced. This ensures that both training and testing sets maintain similar proportions of yes and no classes.

CHAPTER FOUR: Exploratory Data Analysis

Exploratory Data Analysis was conducted to understand the structure of the dataset and the relationship between features and the target variable.

4.1 Target Variable Distribution

The target variable distribution shows a strong class imbalance. Most customers did not subscribe to a term deposit, while only a smaller number subscribed. The test set contained 801 examples of class 0 and only 104 examples of class 1. This means that the dataset is imbalanced, and accuracy alone is not enough to evaluate model performance.

```
print("\nTarget value counts:")
print(df[target_column].value_counts())

print("\nTarget value percentage:")
print(df[target_column].value_counts(normalize=True) * 100)

plt.figure(figsize=(6, 4))
sns.countplot(data=df, x=target_column)
plt.title("Target Variable Distribution")
plt.xlabel("Subscribed to Term Deposit")
plt.ylabel("Count")
plt.savefig("target_distribution.png", dpi=300, bbox_inches="tight")
plt.show()
```

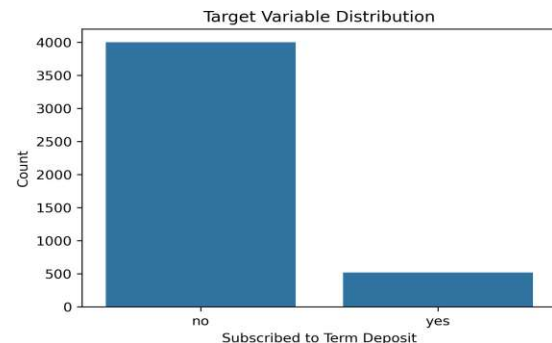


Figure 1. Target variable distribution to the term deposit

In this problem, the positive class yes is very important because it represents customers who subscribed to the term deposit. Therefore, recall and F1-score for class 1 are especially important.

The target distribution (Figure 1) shows that the bank had many unsuccessful campaign outcomes compared with successful ones. This is common in marketing campaigns because only a minority of contacted customers usually accept an offer.

4.2 Age Distribution by Subscription Status

The age distribution plot shows that most customers are between approximately 30 and 60 years old. Both subscribed and non-subscribed customers appear across different age groups. However, the number of non-subscribers is much higher across almost all age ranges.

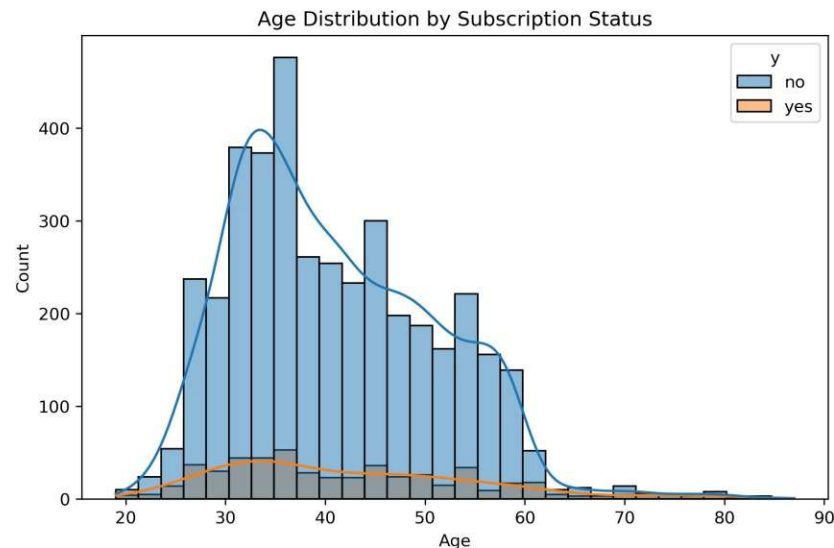


Figure 2. Age distribution of the subscribers

The plot (Figure 2) suggests that age alone may not be a strong predictor of subscription. However, it may still contribute useful information when combined with other features such as job, education, balance, previous campaign outcome, and contact duration.

4.3 Account Balance by Subscription Status

The boxplot (Figure 3) of account balance shows that both subscribed and non-subscribed groups contain many outliers. Some customers have very high account balances, while others have negative balances. The median balance appears relatively low for both groups.

The presence of extreme values suggests that account balance is skewed. Although higher balance may indicate stronger financial capacity, the plot shows that balance alone does not clearly separate subscribers from non-subscribers. Therefore, balance should be used with other predictive variables.

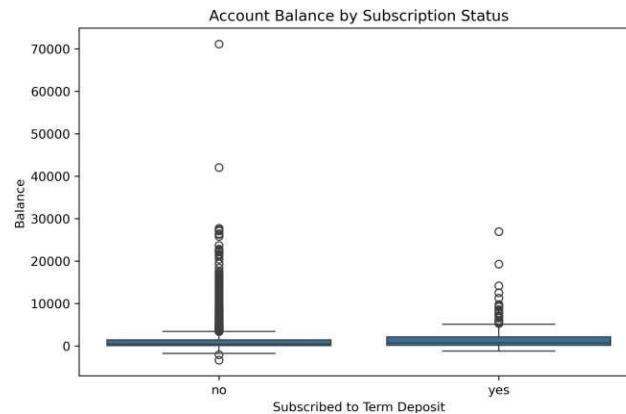


Figure 3. Account balance distribution of all contacted customers

4.4 Number of Campaign Contacts

The campaign contact boxplot (Figure 4) shows the number of times customers were contacted during the campaign. Most customers were contacted only a small number of times, but there are several outliers where customers were contacted many times.

This feature may be important from a business perspective. Too many campaign contacts may reduce customer interest or indicate that previous attempts were unsuccessful. The plot suggests that excessive contacts do not necessarily lead to subscription. This can help banks design more efficient marketing strategies by avoiding too many repeated calls to low-probability customers

4.5 Correlation Heatmap

The correlation heatmap (Figure 5) shows relationships among numerical variables. Most numerical variables have weak correlations with each other. However, *pdays* and *previous* show a moderate positive correlation of approximately 0.58.

This relationship is reasonable because both variables are related to previous marketing contact history. Customers who were contacted previously may have non-zero values for both *pdays* and *previous*. The low correlations among most numerical features suggest that multicollinearity is not a serious problem in the numerical feature set.

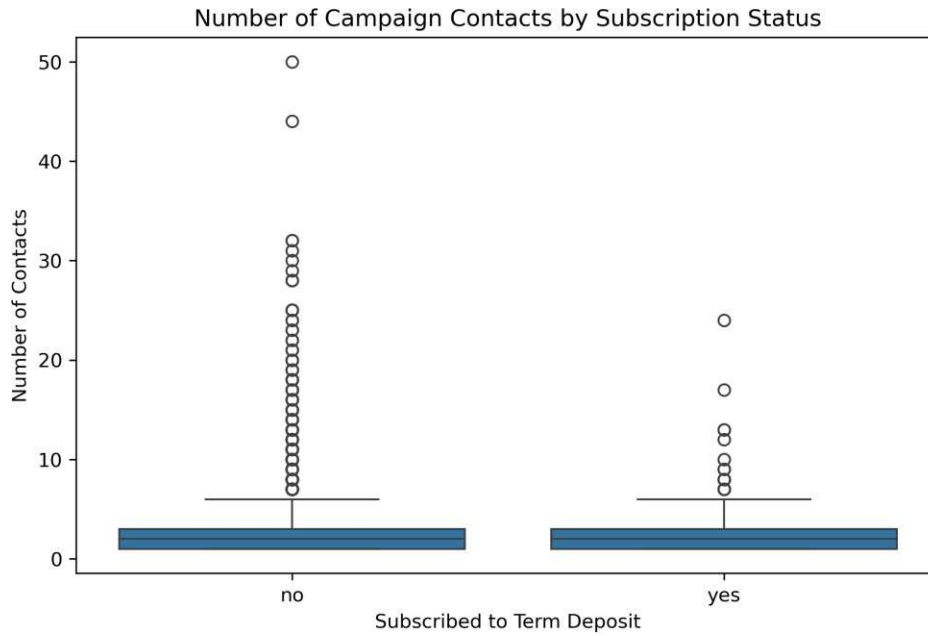


Figure 4. Total number of contact to the customers during the campaign

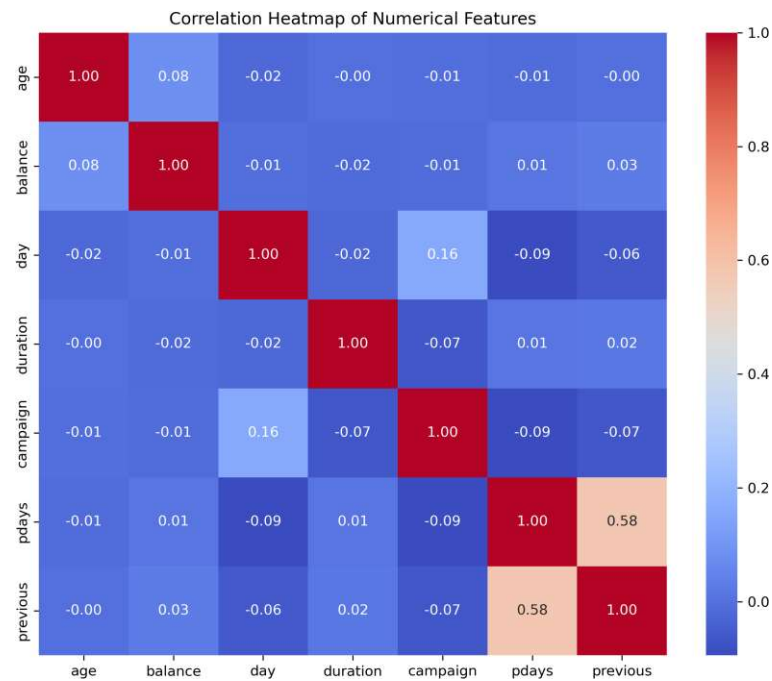


Figure 5. Correlation heatmap of all numerical features

4.6 Education Level and Subscription

The education plot (Figure 6) shows that customers with secondary education form the largest group. Tertiary education customers also form a significant portion. The number of subscribers is lower than non-subscribers in all education groups.

Education may still be useful as a predictor because subscription rates may differ between education levels. Since education has a meaningful order, it was treated as an ordinal variable and encoded accordingly.

4.7 Job Type and Subscription

The job type plot (Figure 7) shows that management, blue-collar, and technician categories have large numbers of customers. Some groups, such as retired and student customers, have smaller total counts but may show different subscription behavior.

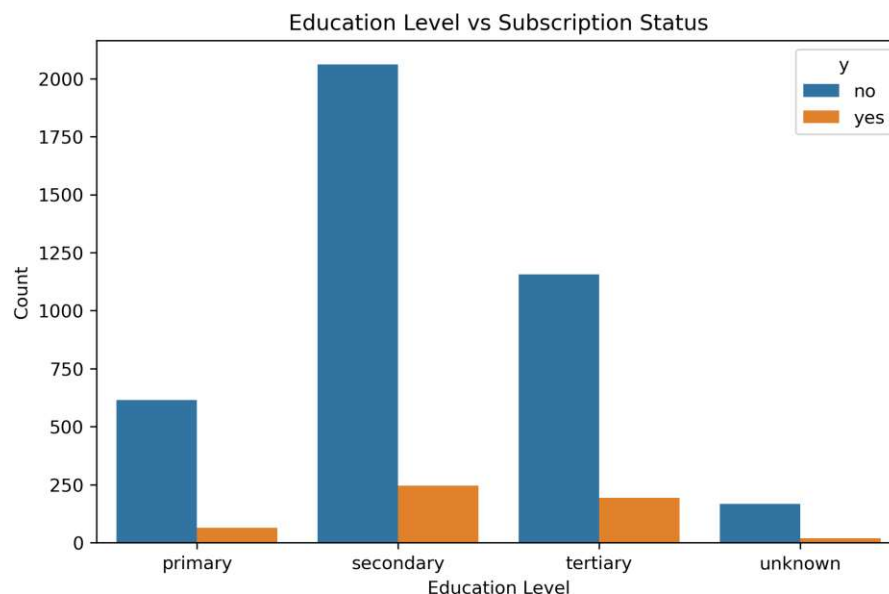


Figure 6. Distribution of education level and subscription status

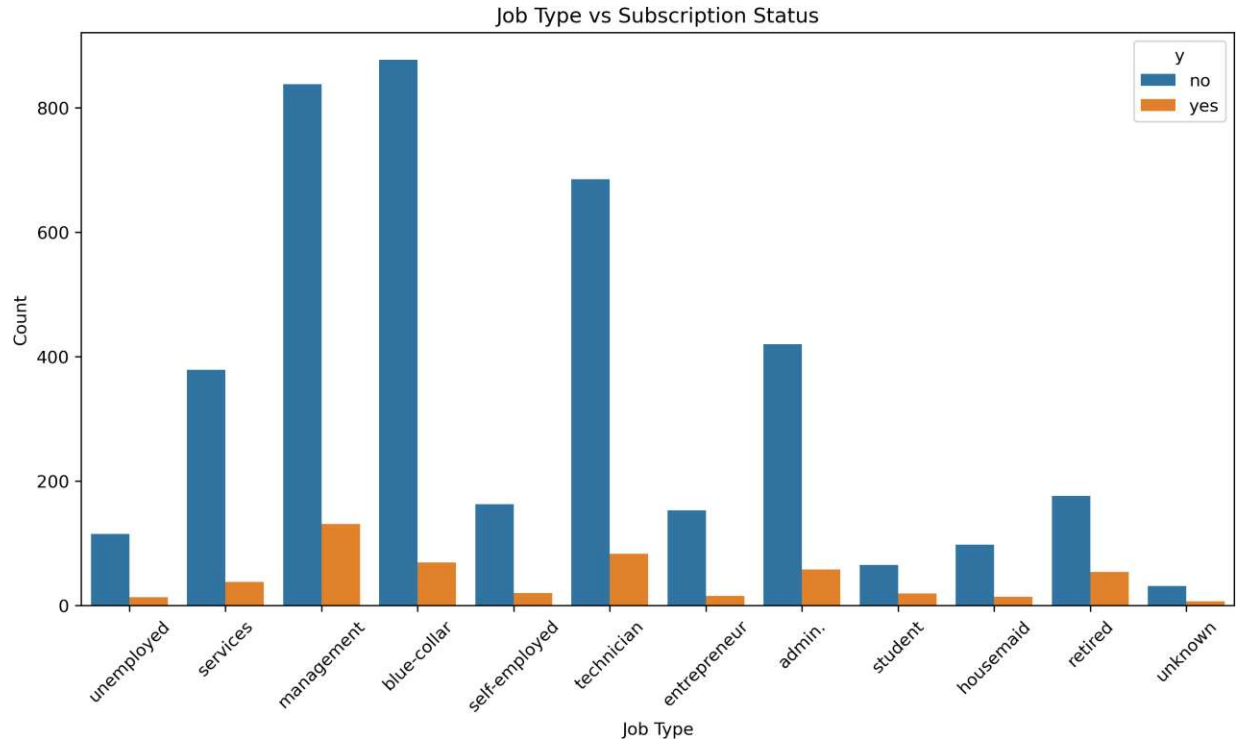


Figure 7. Distribution of job type and subscription status

Job type is important because employment category can reflect income stability, lifestyle, financial goals, and risk preference. Therefore, job type was included in the model using one-hot encoding.

4.8 Marital Status and Subscription

The marital status plot ([Figure 8\(a\)](#)) shows that married customers represent the largest group, followed by single and divorced customers.

Marital status may influence financial decisions because married, single, and divorced customers can have different saving behavior, family responsibilities, and financial planning needs. However, like other demographic features, marital status alone does not clearly separate the two classes.

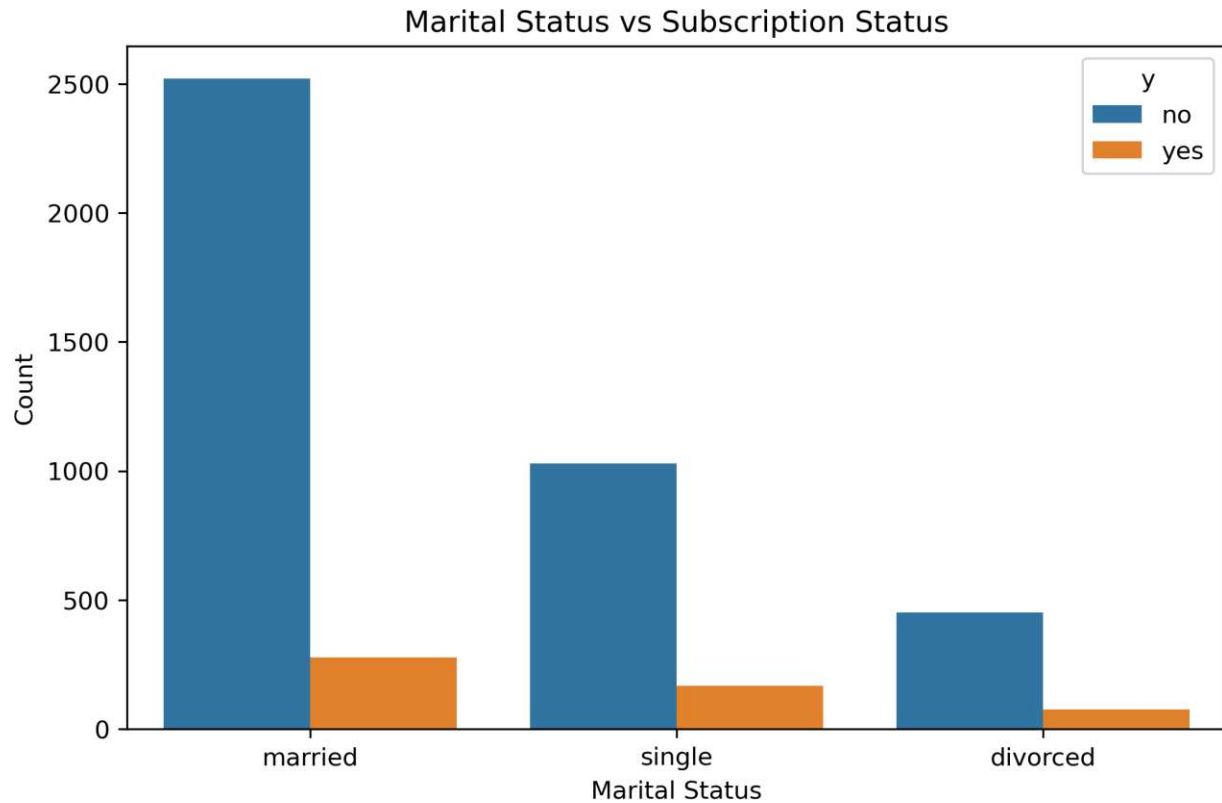


Figure 8(a). Comparison of the marital status people with their subscription status

```
# =====
# Data Cleaning
# =====

df_clean = df.copy()

# Replace "unknown" with NaN
df_clean = df_clean.replace("unknown", np.nan)

print("\nMissing values after replacing 'unknown' with NaN:")
print(df_clean.isnull().sum())

# Separate features and target
X = df_clean.drop(columns=[target_column])
y = df_clean[target_column]

# Encode target variable: yes = 1, no = 0
label_encoder = LabelEncoder()
y_encoded = label_encoder.fit_transform(y)

print("\nTarget encoding:")
for original, encoded in zip(label_encoder.classes_, label_encoder.transform(label_encoder.classes_)):
    print(original, "=", encoded)
```

Figure 8(b). Python code of data cleaning section

CHAPTER FIVE: SVM Classification Model

Support Vector Machine (SVM) ([Cortes, 1995](#)) was used as the first classification model. SVM tries to find an optimal separating boundary between classes. The original SVM method was introduced by Cortes and Vapnik as a supervised learning method for two-group classification problems.

In this project, an SVM model with an RBF kernel was trained using the preprocessed dataset. The model used all relevant numerical and encoded categorical features. The evaluation results were:

Table 3. Support Vector Machine classifier evaluation result

Metric	SVM result
Accuracy	0.895
Precision	0.629
Recall	0.212
F1-score	0.317
ROC-AUC	0.863

The SVM model achieved high accuracy of approximately 89.5%. However, because the dataset is imbalanced, accuracy is not enough to judge model quality. The recall for class 1 was only 0.212, meaning that the model identified only about 21% of actual subscribers. This is a limitation because the business goal is to identify potential customers who are likely to subscribe.

The SVM confusion matrix shows:

Table 4. SVM confusion matrix result

Actual / Predicted	Predicted 0	Predicted 1
--------------------	-------------	-------------

Actual 0	788	13
Actual 1	82	22

The model correctly classified 788 non-subscribers and 22 subscribers. However, it missed 82 actual subscribers by predicting them as non-subscribers (Figure 9 (left)). From a business perspective, these false negatives are important because they represent potential customers that the bank may fail to target.

The SVM ROC curve produced a ROC-AUC of 0.863.

The ROC-AUC score (Figure 9 (right)) indicates that the model has a good ability to rank positive cases higher than negative cases. However, the low recall shows that the default classification threshold does not capture enough positive cases.

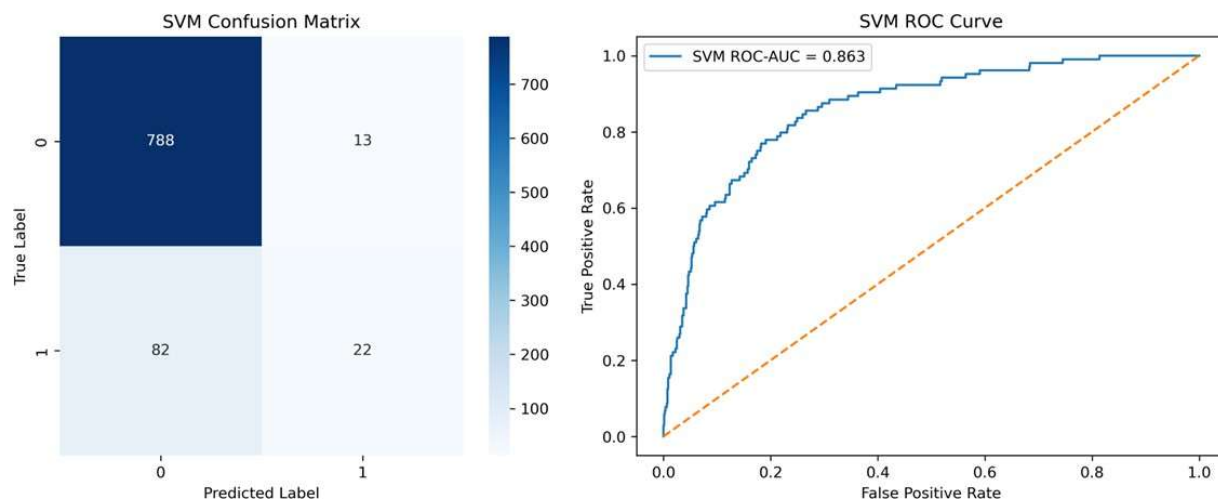


Figure 9. Support vector machine confusion matrix (left) and ROC curve (right)

5. 1 3D SVM Visualization and Decision Boundary

A 3D visualization was created using three numerical features:

- age
- balance

- duration

The full SVM model used all features, but the 3D visualization used only these three features because a three-dimensional decision boundary can only be visualized clearly with three input dimensions.

The 3D plot (Figure 10 (right)) shows the distribution of class 0 and class 1 observations in scaled feature space. A linear SVM decision plane was plotted to show the approximate boundary between the two classes. The visualization shows that the classes overlap significantly. This overlap explains why classification is difficult, especially for the minority class yes.

The decision boundary does not perfectly separate subscribers and non-subscribers. This indicates that the relationship between customer features and subscription behavior is complex and likely non-linear. Therefore, a more flexible algorithm such as XGBoost may perform better.

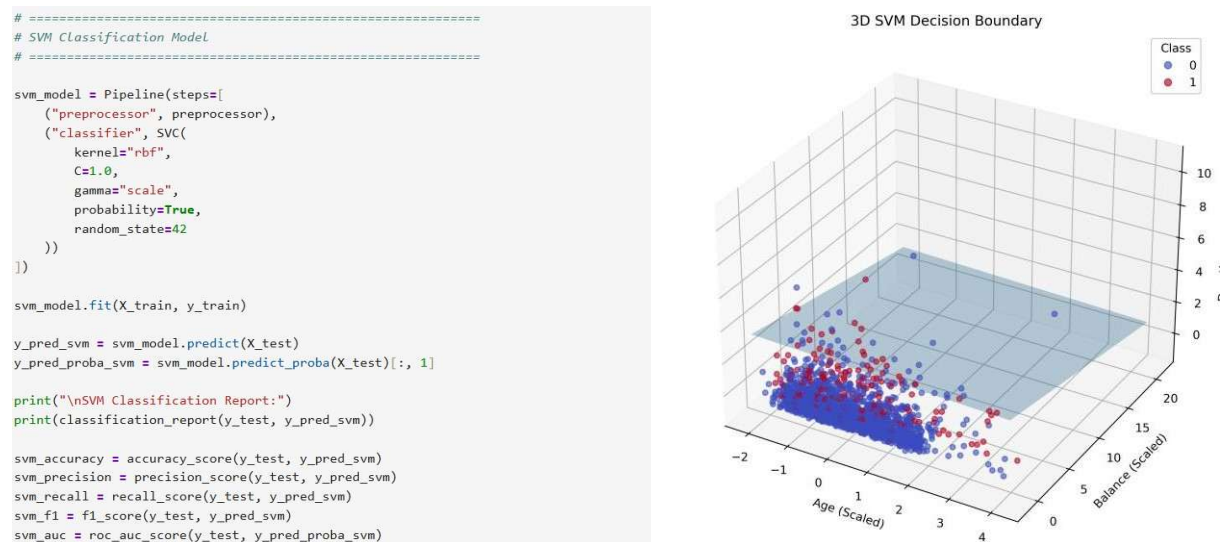


Figure 10. SVM 3D decision boundary and code screenshot

CHAPTER SIX: XGBoost Classification Model

XGBoost ([Chen, August 2016](#)) was used as the second classification model. XGBoost is based on gradient boosted decision trees and is widely used for structured data problems. Chen and Guestrin described XGBoost as a scalable tree boosting system designed for efficient and effective machine learning.

The initial XGBoost model achieved the following results:

Table 5. XGBoost model performance results

Metric	XGBoost result
Accuracy	0.892
Precision	0.545
Recall	0.346
F1-score	0.424
ROC-AUC	0.895

Compared with SVM, the basic XGBoost model had slightly lower accuracy but much better recall and F1-score for the positive class. The recall increased from 0.212 for SVM to 0.346 for XGBoost. The F1-score increased from 0.317 to 0.424.

The XGBoost confusion matrix shows:

Table 6. XGBoost model confusion matrix

Actual / Predicted	Predicted 0	Predicted 1
Actual 0	771	30
Actual 1	68	36

The XGBoost model correctly identified 36 subscribers, compared with only 22 subscribers identified by SVM. Although it produced more false positives than SVM, it reduced false negatives. In marketing, this trade-off may be acceptable because missing potential subscribers can be more costly than contacting some additional customers.

The XGBoost ROC-AUC was 0.895.

This is higher than the SVM ROC-AUC of 0.863, indicating that XGBoost had better overall ranking ability.

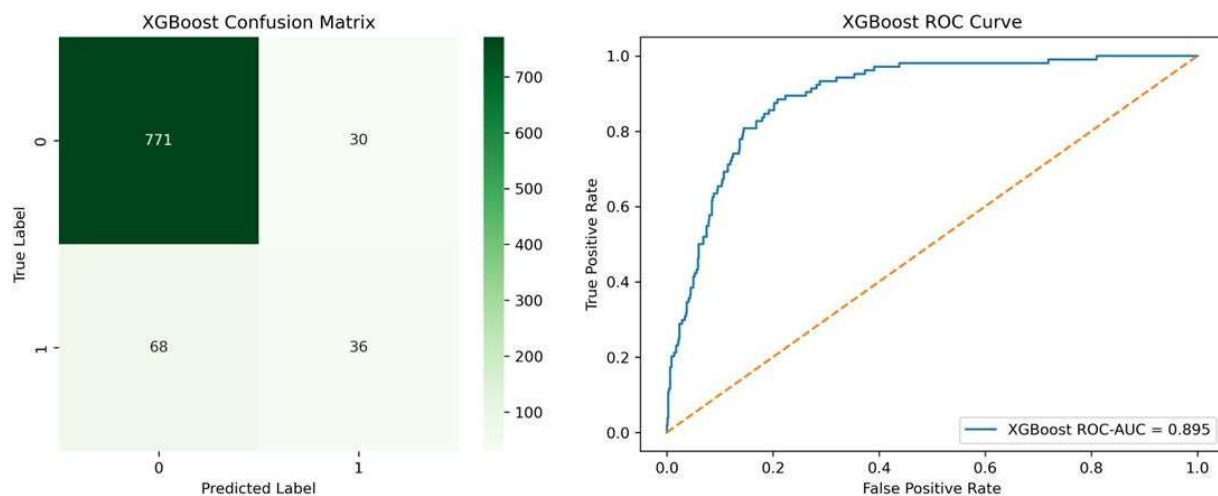


Figure 11. XGBoost confusion matrix (left) and ROC curve (right)

8. XGBoost Hyperparameter Optimization

To improve XGBoost performance, two important hyperparameters were tuned:

- `n_estimators`
- `learning_rate`

The `n_estimators` parameter controls the number of boosting trees. More estimators can improve performance, but too many may increase the risk of overfitting. The `learning_rate` controls how

much each new tree contributes to the final model. A smaller learning rate usually requires more trees, while a larger learning rate allows faster learning but may overfit.

The tested values were:

- `n_estimators` = 50, 100, 200, 300
- `learning_rate` = 0.01, 0.05, 0.10, 0.20

The optimization results showed that the best combination based on lowest error rate was:

- `n_estimators` = 300
- `learning_rate` = 0.05

The best model achieved:

Table 7. Optimized XGBoost performance results

Metric	Optimized XGBoost
Accuracy	0.897
Precision	0.582
Recall	0.375
F1-score	0.456
ROC-AUC	0.901
Error rate	0.103

The error rate visualization shows that the model with learning rate 0.05 and 300 estimators produced the lowest error rate.

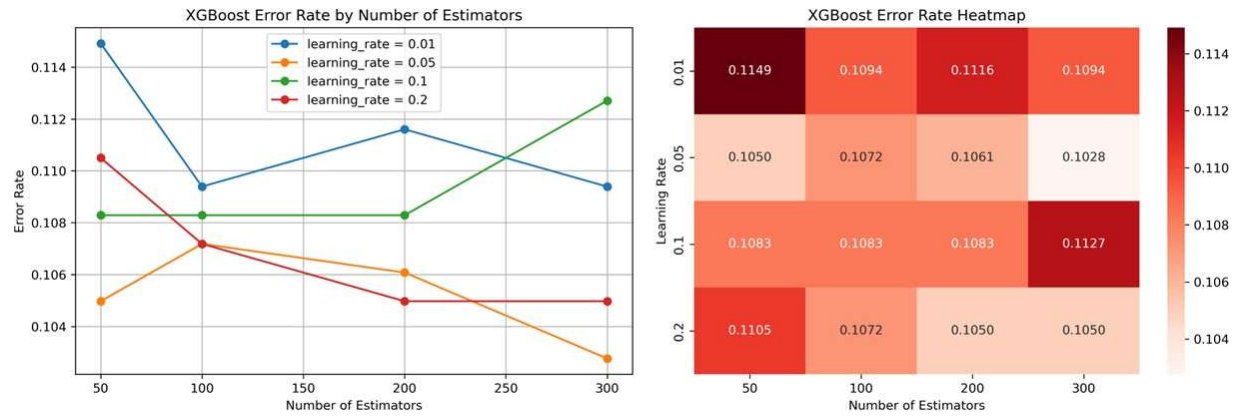


Figure 12. XGBoost error rate by estimators (left) and heatmap (right)

The F1-score visualization shows that higher numbers of estimators generally improved F1-score, especially when combined with learning rates 0.05 or 0.20.

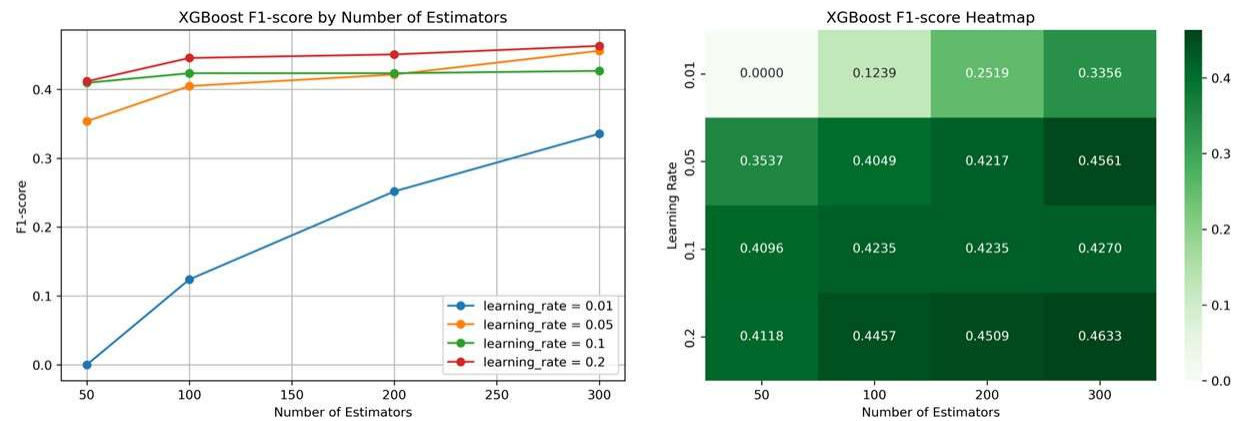


Figure 13 (a). XGBoost f1 score error rate by estimator (left) and f1 score heatmap (right)

Although learning rate 0.20 produced the highest F1-score in one setting, the lowest error rate was achieved by learning rate 0.05 with 300 estimators. Therefore, this combination was selected as the optimized model.

```

# =====
# XGBoost Hyperparameter Tuning
# =====

n_estimators_values = [50, 100, 200, 300]
learning_rate_values = [0.01, 0.05, 0.1, 0.2]

results = []

for n_est in n_estimators_values:
    for lr in learning_rate_values:
        model = Pipeline(steps=[
            ("preprocessor", preprocessor),
            ("classifier", XGBClassifier(
                n_estimators=n_est,
                learning_rate=lr,
                max_depth=4,
                objective="binary:logistic",
                eval_metric="logloss",
                random_states=42
            ))
        ])
        model.fit(X_train, y_train)

        y_pred = model.predict(X_test)
        y_pred_proba = model.predict_proba(X_test)[:, 1]

        accuracy = accuracy_score(y_test, y_pred)
        precision = precision_score(y_test, y_pred)
        recall = recall_score(y_test, y_pred)
        f1 = f1_score(y_test, y_pred)
        auc = roc_auc_score(y_test, y_pred_proba)
        error_rate = 1 - accuracy

        results.append({
            "n_estimators": n_est,
            "learning_rate": lr,
            "accuracy": accuracy,
            "precision": precision,
            "recall": recall,
            "f1_score": f1,
            "roc_auc": auc,
            "error_rate": error_rate
        })

results_df = pd.DataFrame(results)

print("\nXGBoost Tuning Results:")
print(results_df)

best_result = results_df.sort_values(by="error_rate").iloc[0]

print("\nBest Hyperparameter Combination:")
print(best_result)

```

Figure 13(b): XGBoost hyperparameter tuning Python code screenshot

CHAPTER SEVEN: Model Comparison and Feature Importance

The final comparison among SVM, basic XGBoost, and optimized XGBoost is shown below:

Table 8. Performance comparison table of SVM, XGBoost, and Optimized XGBoost models

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
SVM	0.895	0.629	0.212	0.317	0.863
XGBoost	0.892	0.545	0.346	0.424	0.895
Optimized XGBoost	0.897	0.582	0.375	0.456	0.901

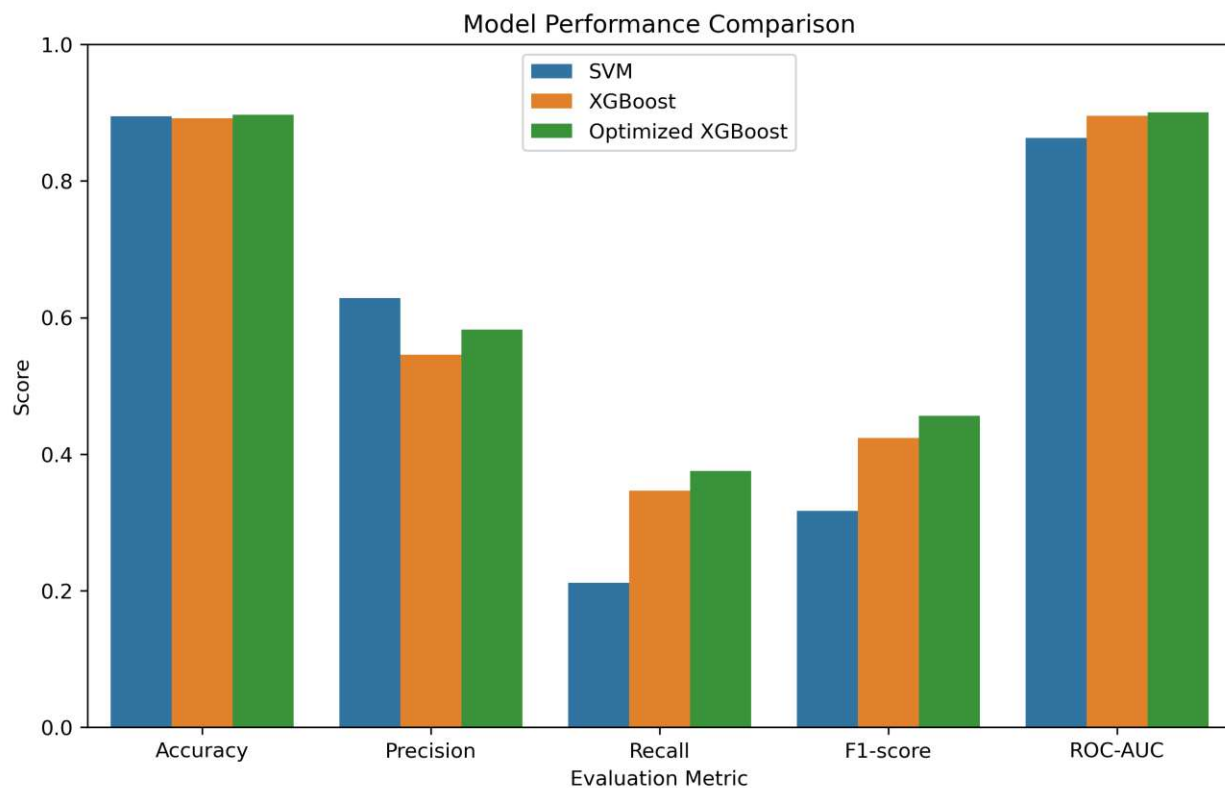


Figure 14. All model performance comparison

The comparison shows that SVM had the highest precision but the lowest recall. This means that when SVM predicted a customer as a subscriber, it was relatively careful, but it missed many

actual subscribers. XGBoost had better recall and F1-score, meaning it was more effective at detecting the positive class.

The optimized XGBoost model performed best overall. It achieved the highest accuracy, highest recall, highest F1-score, and highest ROC-AUC. Therefore, optimized XGBoost is the most suitable model for this business problem.

Feature Importance Analysis

The XGBoost feature importance plot shows the most influential predictors in the model.

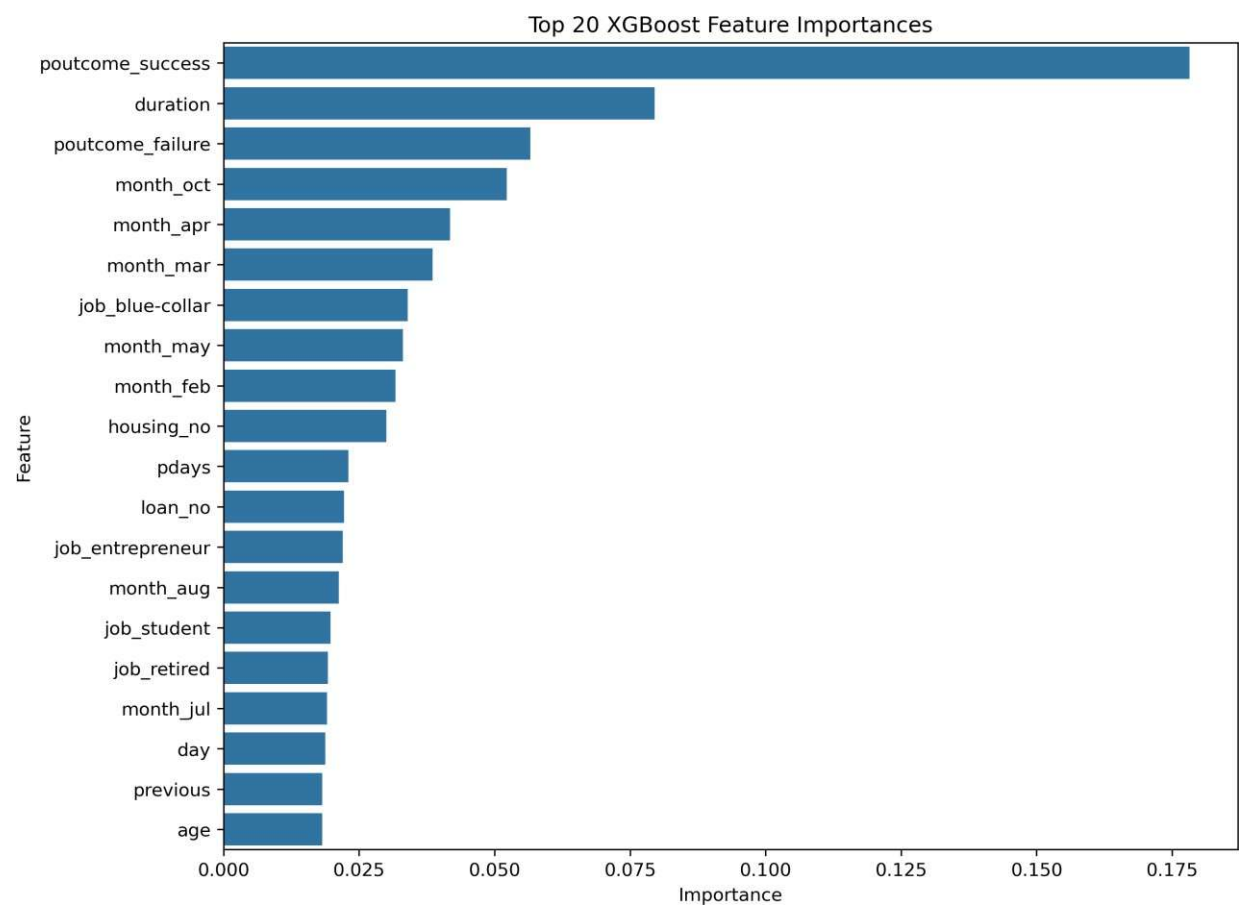


Figure 15. XGBoost feature importances visualization

The top features were:

Feature	Interpretation
---------	----------------

poutcome_success	Previous campaign success strongly predicts current subscription
duration	Longer call duration is associated with higher subscription likelihood
poutcome_failure	Previous failure also provides useful campaign history
month_oct, month_apr, month_mar	Contact month affects campaign success
job_blue-collar	Job category influences subscription probability
housing_no	Customers without housing loans may behave differently
pdays	Time since previous contact is important
previous	Previous campaign contact history matters
age	Age contributes some predictive information

The most important feature was `poutcome_success`, meaning that customers who had a successful outcome in a previous campaign were more likely to subscribe again. This makes strong business sense because previous positive engagement is often a strong indicator of future interest.

The second most important feature was `duration`. This suggests that longer conversations may indicate higher customer interest. However, in a real deployment, `duration` should be used carefully because it is only known after the call has already happened. If the goal is to predict before calling the customer, `duration` may not be available and should be removed from the model.

The importance of contact months such as October, April, and March suggests that campaign timing may influence customer response. This can help banks plan campaigns during more effective periods.

CHAPTER EIGHT: Business Implications

The findings have several practical implications for bank marketing strategy.

First, the bank should not rely on random customer targeting. The dataset shows that only a minority of customers subscribed to a term deposit. Therefore, contacting all customers equally may waste resources. A predictive model can help prioritize customers who are more likely to subscribe.

Second, previous campaign outcome is highly important. Customers who responded positively in the past should be considered high-priority leads. The bank can use previous campaign history to design more personalized marketing strategies.

Third, the number of campaign contacts should be controlled. The EDA showed that many contacts do not necessarily lead to successful subscription. Excessive contacts may reduce customer satisfaction and waste employee time.

Fourth, XGBoost is more suitable than SVM for this problem. Although SVM achieved good accuracy, it failed to detect many actual subscribers. Optimized XGBoost improved recall and F1-score for the positive class, making it more useful for business decision making.

Finally, the model can support data-driven campaign planning. Marketing teams can use predicted probabilities to rank customers and focus on those with the highest likelihood of subscription. This can improve campaign efficiency, increase conversion rates, and reduce operational costs.

CHAPTER NINE: CONCLUDING REMARKS

This report applied machine learning and optimization techniques to predict customer subscription to a bank term deposit using the UCI Bank Marketing dataset. The project followed a complete analytical workflow, including data understanding, preprocessing, encoding, exploratory data analysis, model training, evaluation, visualization, and hyperparameter tuning.

The dataset contained numerical, categorical, binary categorical, and ordinal variables. Numerical features were standardized, categorical variables were one-hot encoded, and the ordinal education variable was ordinal encoded. The target variable was converted into binary form, where no = 0 and yes = 1.

Exploratory analysis showed that the dataset was imbalanced, with many more non-subscribers than subscribers. Although several features such as age, balance, education, job type, marital status, campaign contacts, and previous campaign information were useful, no single feature clearly separated the two classes.

The SVM model achieved good accuracy of 0.895 and ROC-AUC of 0.863, but it performed poorly in identifying actual subscribers, with only 22 out of 104 subscribers correctly classified. In contrast, XGBoost performed better for the positive class. After tuning, the optimized XGBoost model achieved the best overall performance, with accuracy of 0.897, recall of 0.375, F1-score of 0.456, and ROC-AUC of 0.901.

The most important features were previous campaign success, call duration, previous campaign failure, contact month, job type, housing loan status, previous contact days, and age. Overall, optimized XGBoost was the most suitable model for this problem. It can help the bank identify customers who are more likely to subscribe, reduce unnecessary marketing calls, improve campaign efficiency, and support data-driven decision making.

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